

2016: selected excess 2 at a 69 MeV signal hypothesis

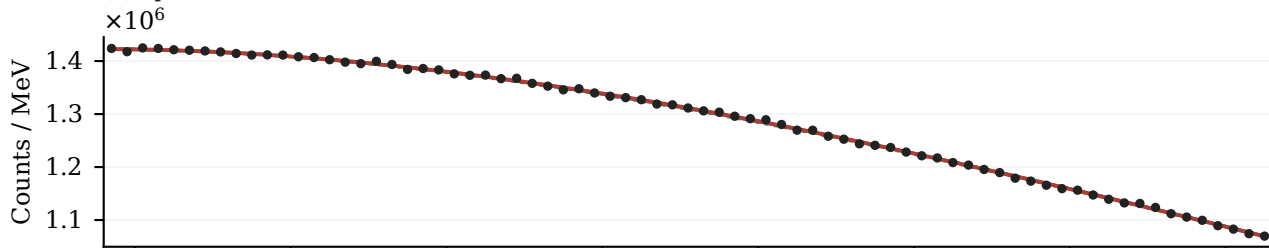
Neighboring signal-MC template; window $-3.5u$ to $+3.5u$ | $c = 68.837$ MeV, $s = 2.701$ MeV

Signed full-selected yield = $16,791 \pm 7,371$; local asymptotic $Z = 2.28 \sigma$, $p = 0.0114$

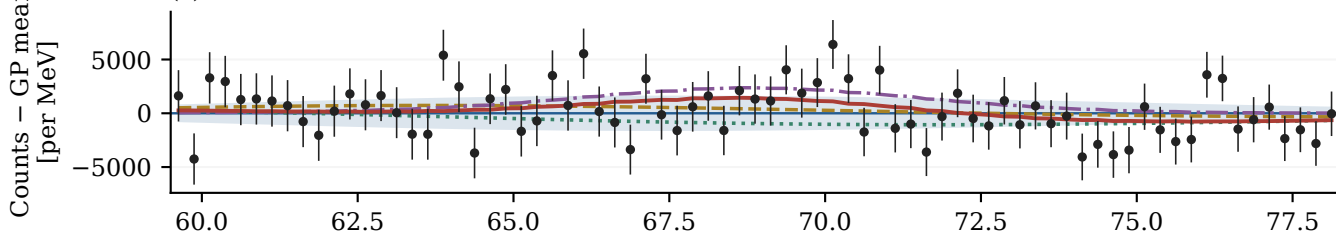
Fixed-source null toys: local rank $p = 0.00599$ (5 / 1000 exceedances); count 95% interval [0.00163, 0.0116]

- Observed counts
- ⋯ Background in signed S+B fit
- ⬜ Fit interval: excluded from GP training
- ⬜ GP mean and ± 1 SD constraint
- Signed signal + background
- Search mass hypothesis
- Background-only fit
- Signed signal (middle panel)
- Signal core center

(a) Spectrum in the fit interval



(b) Residuals relative to the GP mean



(c) Standardized residuals and excluded fit interval

